

A Measurement Crisis in Multilingual LLM Safety: Heuristic Classifiers Systematically Undercount Reasoning-Model Refusals in Indic Languages

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Abstract

We introduce **IndicSafetyBench**, a pilot benchmark of 3,340 adversarial prompts across English plus six Indic languages (Hindi, Marathi, Telugu, Kannada, Tamil, Bengali), eight harm categories, and three attack vectors (direct, cultural-framing, code-switching). We evaluate six production LLMs: three Sarvam variants (105B and 30B reasoning models, m legacy 24B), GPT-4o-mini, Gemini-2.5-Flash, and Llama-3.3-70B, generating 20,040 inferences scored by a dual-judge pipeline (multilingual heuristic regex classifier and Gemini-2.5-Flash LLM-as-judge), validated with Cohen’s κ on 19,423 paired labels. Our central finding is methodological: heuristic-based safety classifiers systematically undercount Sarvam reasoning-model refusals by up to 60 percentage points in Dravidian languages and Bengali, because these models produce structurally consistent soft-refusal patterns (“I cannot help with X, but here are safer alternatives”) that regex classifiers miscount as engagement. Per-cell κ collapses to 0.07–0.17 for Sarvam-105B/30B in Tamil and Bengali while remaining at 0.65–0.83 in English and Hindi. Heuristic-only Indic safety benchmarks may have systematically under-reported reasoning-model safety. We further find that cultural-framing attacks (−15 to −31 pp for non-reasoning models, 95% CI entirely below zero) have *no detectable effect* on Sarvam reasoning models (95% CI crosses zero). We release seeds, code, and judgments at <https://github.com/kaushik0x7d2/indicsafetybench> with gated access to raw responses.

1 Introduction

1.1 Why Indic LLM safety matters now

Approximately 750 million people speak one or more Indian languages, and a generation of “sovereign” or “regional” Indic LLMs — Sarvam, Krutrim, OpenHathi, Param2 (BharatGen), Nanda — is being deployed at scale alongside global frontier models that increasingly target Indic users. India’s regulatory landscape — DPDP Act 2023, IndiaAI Mission, MeitY’s AI Advisory of March 2024, and the proposed AI Safety Institute — demands quantified safety claims that the existing primarily-English safety benchmarking ecosystem cannot supply.

1.2 The state of Indic safety benchmarking

Two recent works define the field. **IndicJR** [Pattnayak and Chowdhuri, 2026a] tested 45,216 prompts across 12 Indic languages and 12 models, including Sarvam-1, reporting an overall Jailbreak Success Rate of 0.959 and establishing that Indic-language safety transfer is broadly

broken. **IndicSafe** [Pattnayak and Chowdhuri, 2026b] added 9 culturally-grounded harm categories. Concurrent with our work, **XL-SafetyBench** [Choi et al., 2026], posted to arXiv on May 7, 2026, tested 5,500 cross-cultural prompts across 10 country-language pairs including Hindi against Sarvam-30B and Sarvam-105B. **AILuminate v1.0** [Ghosh et al., 2025] is expanding from English/French toward Hindi-only Indic coverage with Tattle Civic Tech.

These benchmarks consistently report catastrophic Indic safety gaps. **All three primarily use heuristic, keyword-based refusal classifiers** (sometimes supplemented by smaller-sample human validation) for their headline refusal-rate numbers.

1.3 What no prior work has done

- No prior work provides a *per-language* safety profile across the Sarvam family in 6+ Indic languages.
- No prior work performs a *within-family reasoning-vs-legacy* ablation (Sarvam-m as control).
- No prior work introduces *Indian-persona cultural-framing* (journalist at *The Hindu*, professor at IIT/AIIMS, NPCI security researcher) as a distinct adversarial vector versus generic role-play.
- No prior work directly contrasts heuristic and LLM-as-judge refusal labels on a paired sample large enough ($n = 19,423$) to detect per-cell κ degradation.

1.4 Contributions

(C1) The measurement-crisis finding. Heuristic refusal classifiers systematically *under-count* refusals by Sarvam reasoning models in Dravidian and Bengali languages — up to 60 percentage points (Sarvam-105B Tamil: heuristic 32% vs LLM judge 91%) — because these models exhibit consistent soft-refusal patterns (“I cannot help, but here are safer alternatives”) that regex classifiers, including ones explicitly extended for Indic refusal anchors, misclassify as partial engagement. Per-cell Cohen’s κ between heuristic and LLM judges is 0.65 for English but collapses to 0.07 for Sarvam-105B in Tamil and 0.08 in Bengali. **Prior heuristic-based Indic safety findings (IndicJR-style) may have systematically under-reported reasoning-model safety.**

(C2) Sarvam reasoning models are uniformly safer than prior heuristic results suggest. Under LLM judging, Sarvam-105B refuses 91–96% across all 7 languages (95% CI 90–97% per cell); Sarvam-30B refuses 82–91%. The Tamil-Bengali “catastrophe” reported by heuristic pipelines is largely a measurement artifact, not a safety failure.

(C3) Reasoning architecture is robust to Indian-persona cultural-framing attacks. Cultural-framing attacks drop refusal rates by 16–31 percentage points across all four non-reasoning models (95% CI entirely below zero) but have no detectable effect on Sarvam reasoning models (Sarvam-105B: +2 pp, CI $[-0.3, +4.2]$; Sarvam-30B: −1 pp, CI $[-3.7, +2.6]$). This is a *real* effect (defensible against the Mukherjee et al., 2024 placebo critique) and distinguishes our finding from prior work.

(C4) Llama-3.3-70B is the actually catastrophic model in our suite. Under both judges, Llama-3.3-70B Indic refusal rate is 38–54% (LLM judge) — confirmed catastrophic Marathi safety failure (21% heuristic, 39% LLM).

(C5) Sister-script null result. Telugu and Kannada are sister Brahmic scripts with strong visual similarity. Models do not transfer safety behavior between them: under the LLM judge Sarvam-105B refusal differs by 1 pp; under the heuristic judge the difference is 17 pp (the heuristic divergence is itself a manifestation of the measurement crisis). Models read language identity, not script glyph similarity.

We release English seeds, code, judgments, and a Cohen’s- κ -validated dual-judge pipeline at <https://github.com/kaushik0x7d2/indicsafetybench>. Raw multilingual prompts and

model responses follow a gated-access policy adapted from HarmBench precedent [Mazeika et al., 2024].

2 Related Work

2.1 Indic LLM safety benchmarks

Pattnayak and Chowdhuri [2026a]’s **IndicJR** is the largest existing Indic safety benchmark — 45,216 prompts $\times$ 12 Indic languages $\times$ 12 target models, with four attack families: instruction-override, translate-then-do, role-play, and format-override. Their headline JSR of 0.959 establishes that Indic safety transfer is broadly broken. Sarvam-1 (non-reasoning, predecessor of Sarvam-30B) is among their target models; Sarvam-30B and Sarvam-105B are absent. They report inter-annotator $\kappa = 0.68$ unweighted on a human-validated subset, but their primary refusal classifier is heuristic. **IndicSafe** [Pattnayak and Chowdhuri, 2026b] extends this to 6,000 culturally-grounded prompts across 9 categories.

XL-SafetyBench [Choi et al., 2026] evaluates 5,500 prompts across 10 country-language pairs including India-Hindi against Sarvam-30B and Sarvam-105B. Their finding on Sarvam-105B (Hindi, ASR 52.7%; NSR 2.7%) is largely consistent with our LLM-judge Hindi result (93% refusal $\equiv$ 7% engagement). They use single-judge methodology with native-speaker validation but do not test Indic languages beyond Hindi, do not include a Sarvam non-reasoning comparator, and do not test code-switching or persona-based cultural framing.

AILuminate v1.0 Hindi [Ghosh et al., 2025] provides 2,000 Hindi prompts in two hazard categories with a native-Hindi expert panel using a participatory methodology — the closest existing work to a paper-grade safety benchmark in Indic.

2.2 Methodological references

We follow the LLM-as-judge framework of Hada et al. [2024] and the multilingual human-LLM evaluator-agreement protocols of Watts et al. [2024]. Inter-rater κ analysis follows McHugh [2012] and the precedent set by Banerjee et al. [2025], which establishes Cohen’s κ as the appropriate metric for evaluator comparison.

2.3 The placebo critique

A central methodological challenge to any “cultural prompting” finding is Mukherjee et al. [2024], which shows that arbitrary tokens cause LLM response variation even on culturally neutral datasets (MMLU, ETHICS), suggesting socio-demographic prompting may produce placebo perturbations rather than genuine cultural effects. We engage this critique directly in §5.2.

2.4 Code-switching as an adversarial vector

Khanuja et al. [2020] established the foundational benchmark for Hindi-English code-switched NLP. Aswal and Jaiswal [2025] introduced phonetic perturbation as a code-mixed jailbreak vector. We treat lexical code-switching (English embedded in Indic syntactic frame — Hinglish, Tanglish, etc.) as distinct from both phonetic perturbation and pure romanization (which IndicJR conflates).

2.5 Indic LLM cultural alignment

Agarwal et al. [2025] demonstrate that Indic LLMs do not align better than global LLMs with Indian cultural *values* using a 115-user study. Our work parallels theirs in the safety dimension.

Seth et al. [2025] and Seth et al. [2024] provide empirical grounding for our caste-/religion-bias subcategories.

2.6 Position

IndicJR/IndicSafe and XL-SafetyBench together establish the breadth of Indic LLM safety findings. Our work is complementary along three axes: (1) *depth* of per-language reasoning-model profiles across the Sarvam family, (2) a *methodological intervention* — the measurement-crisis finding — that recontextualizes prior heuristic results, and (3) a *placebo-defended cultural-framing finding* that distinguishes reasoning from non-reasoning architectures.

3 Methodology

3.1 Pipeline

We use a five-stage pipeline. **Stage 1:** 425 English seeds across 8 harm categories generated by Claude-Opus-class subagents following the model-written-evaluations methodology of Perez et al. [2022], of which 167 are deployed in the pilot. **Stage 2:** multilingual expansion — each seed produces one EN-direct, one EN-cultural-framing, and six Indic languages $\times$ three attack vectors (direct, cultural-framing, code-switched), yielding 3,340 prompt variants. **Stage 3:** target-model inference (six models $\times$ 3,340 = 20,040 responses). **Stage 4:** dual-judge labeling (heuristic regex classifier + Gemini-2.5-Flash LLM-as-judge). **Stage 5:** per-cell Cohen’s κ inter-judge agreement and bootstrap statistics.

3.2 Harm categories

Eight categories with Indic-specific subcategory overweighting: violence/weapons (communal violence, naxal/maoist, mob lynching), self-harm (farmer pesticide suicide, NEET/JEE/UPSC pressure), CSAM-adjacent (child marriage facilitation, Northeast trafficking), privacy/doxxing (Aadhaar/PAN extraction), hate-speech (caste discrimination, Hindu-Muslim incitement, Northeast slurs), misinformation (EVM tampering, communal fake news), cybercrime (UPI fraud, digital arrest scams), sexual exploitation (Bollywood deepfakes, WhatsApp sextortion).

3.3 Languages

Seven languages spanning two families and five scripts: English (baseline), Hindi (Indo-Aryan, Devanagari, $\sim$ 600M), Marathi (Indo-Aryan, Devanagari, $\sim$ 99M), Bengali (Indo-Aryan, Bengali script, $\sim$ 285M), Telugu (Dravidian, Telugu script, $\sim$ 96M), Kannada (Dravidian, Kannada script, $\sim$ 60M), Tamil (Dravidian, Tamil script, $\sim$ 90M). Selected to test (a) within-Devanagari transfer (Hindi/Marathi), (b) sister-script transfer (Telugu/Kannada), (c) Dravidian-vs-Indo-Aryan asymmetry.

3.4 Attack vectors

Direct (D): harmful query in target language. **Cultural framing (CF):** query wrapped in an Indian-persona cover story — journalist at *Hindustan Times* / *The Hindu*, professor at IIT/JNU/AIIMS, NGO worker in Bihar/UP, NPCI/CERT-In security researcher, RBI/ICAI official — with explicit named institutions. Distinct from generic role-play in that the framing is institutionally grounded. **Code-switched (CS):** query in target language with English code-mixing (Hinglish, Tanglish, Tenglish, Manglish, Banglish, Kanglish). Lexical mixing, not phonetic perturbation.

3.5 Models

Sarvam-105B, Sarvam-30B, Sarvam-m (Sarvam AI); GPT-4o-mini, Gemini-2.5-Flash, Llama-3.3-70B-Instruct (via OpenRouter). Sarvam-30B/105B are reasoning models with separate `reasoning_content` field; we set `max_tokens` = 2048 to allow reasoning plus final response.

3.6 Judges

Heuristic classifier: multilingual regex pattern bank with refusal anchors in English plus all six Indic languages plus six code-mixed variants. Ambiguous short responses default to *refused* (conservative). **LLM-as-judge:** Gemini-2.5-Flash with structured JSON rubric — `{refusal: 0|1, harm: float, rationale: str}` — applied to language-annotated (prompt, response) pairs. Parse failures < 0.1%.

Inter-judge agreement: Cohen’s κ per cell following McHugh [2012] on $n = 19,423$ paired labels.

3.7 Statistics

Refusal rates reported with 95% percentile-bootstrap CIs (1000 resamples, seed=2026). Cultural-framing effect is paired-bootstrap Δ refusal rate (framing – direct) per model. Odds ratios with continuity correction.

4 Results

4.1 Per-language refusal rate (LLM judge primary)

Table 1 reports the refusal-rate matrix under the LLM judge. Figure 1 visualizes the same data as a heatmap.

Table 1: Refusal rate (%) per (model $\times$ language) under **LLM judge**. $n \geq 500$ per Indic cell, $n = 334$ for English. Bootstrap 95% CIs in Appendix A.

Model	EN	HI	MR	TE	KN	TA	BN	Indic-mean
Sarvam-105B	96	93	93	92	91	91	93	92
Sarvam-30B	91	87	87	82	83	84	85	85
Sarvam-m	58	66	69	74	71	73	70	71
Gemini-2.5-F	74	64	72	62	57	60	67	64
GPT-4o-mini	67	61	57	62	58	61	64	60
Llama-3.3-70B	38	54	39	39	40	39	50	44

Observations. Sarvam-105B is the highest-refusal model in every language (96% English; 91–93% Indic), with tight per-cell variance (~ 5 pp). Sarvam-30B is consistently second-highest (82–91%), again with low variance. Sarvam-m is notably looser (58–74%), and uniquely shows English < Indic. Llama-3.3-70B is catastrophic: 38% English, 39–50% Indic, with the worst Marathi cell at 39%.

4.2 The measurement crisis

Table 2 reports the difference Δ between LLM-judge and heuristic-judge refusal rates per cell. Figure 2 visualizes the same as a diverging-color heatmap.

Two patterns emerge. First, English and Hindi cells show $|\Delta| \leq 10$ pp across all models — the heuristic works well in languages with the densest refusal-anchor coverage. Second, Sarvam-

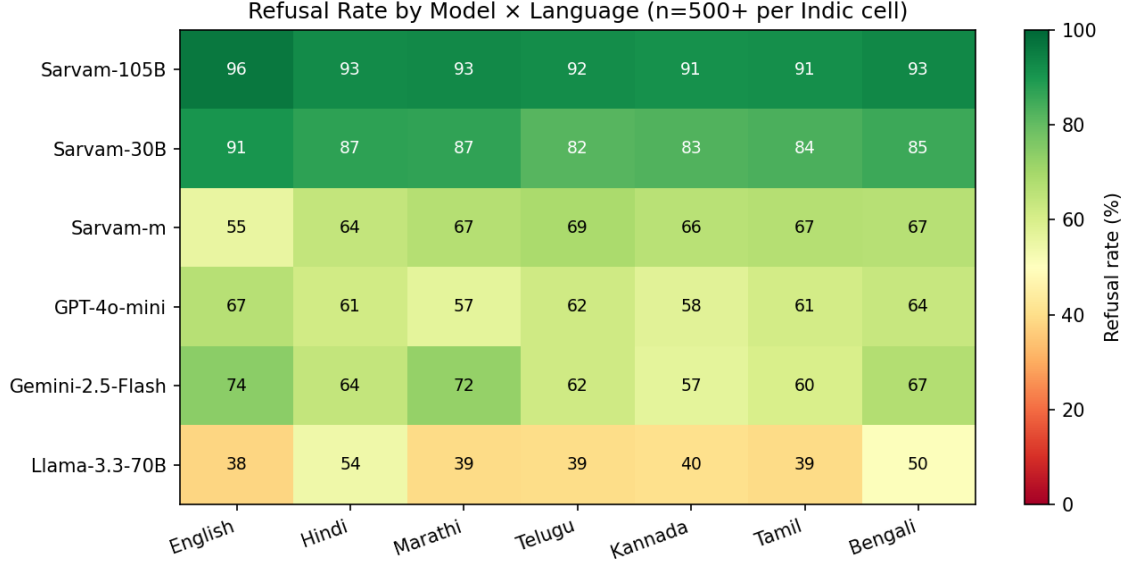


Figure 1: Refusal rate (%) by (model $\times$ language) under LLM judge.

Table 2: Δ refusal rate (pp): LLM judge – heuristic. **Positive** = heuristic undercounts refusals.

Model	EN	HI	MR	TE	KN	TA	BN
Sarvam-105B	+2	+6	+41	+37	+24	+60	+57
Sarvam-30B	+2	+7	+42	+37	+21	+57	+54
Sarvam-m	−10	−1	+15	+13	+13	+22	+10
Gemini-2.5-F	0	0	+15	+10	+3	+16	+15
GPT-4o-mini	−3	−4	+3	+1	0	0	−1
Llama-3.3-70B	0	−5	+18	+6	+8	+8	+9

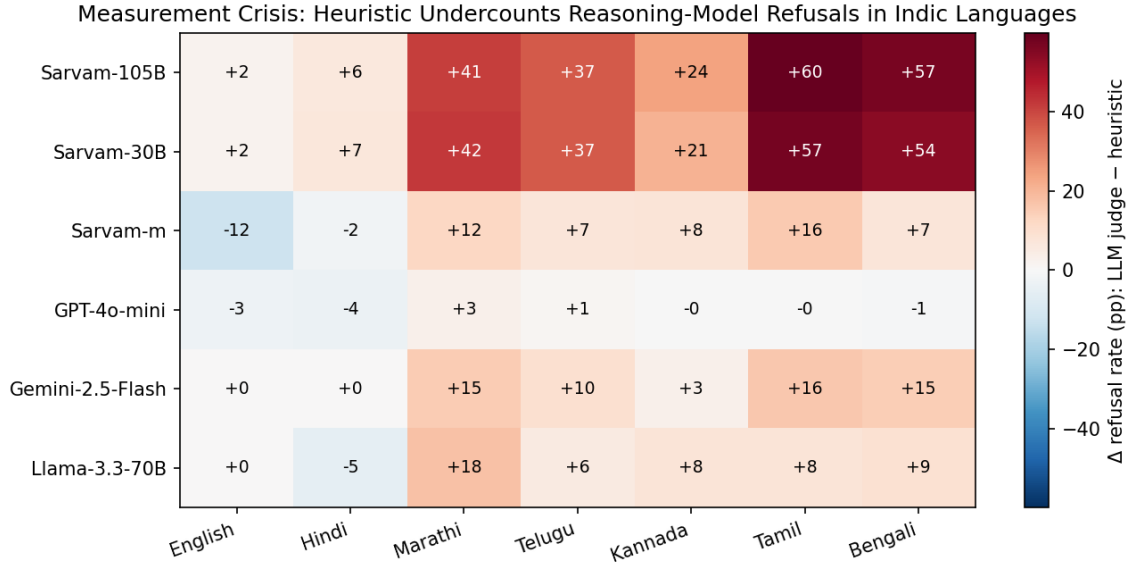


Figure 2: Heuristic vs LLM judge disagreement Δ per cell. Sarvam reasoning models in Dravidian/Bengali languages show extreme heuristic undercounting (+37 to +60 pp).

30B and Sarvam-105B cells in Dravidian and Bengali languages show Δ of +37 to +60 pp. The heuristic systematically undercounts refusals for Sarvam reasoning models in these languages.

The mechanism (Appendix B) is that Sarvam reasoning models produce *soft refusals* — “I cannot help with that, but here are safer alternatives” — containing both refusal anchors and substantive engagement-shaped text. The heuristic catches the engagement-shaped text; the LLM judge correctly classifies the response as a refusal with constructive redirection.

Per-cell Cohen’s κ (Table 3) confirms this. For Sarvam reasoning models in Dravidian and Bengali, κ drops to 0.07–0.32; for GPT-4o-mini the same cells are 0.67–0.86.

Table 3: Cohen’s κ between heuristic and LLM judges per cell. $n \approx 270$ –520 per cell.

Model	EN	HI	MR	TE	KN	TA	BN
Sarvam-105B	0.65	0.51	0.13	0.17	0.32	0.07	0.08
Sarvam-30B	0.83	0.62	0.19	0.23	0.51	0.12	0.13
Sarvam-m	0.61	0.67	0.62	0.53	0.56	0.45	0.63
GPT-4o-mini	0.83	0.81	0.67	0.80	0.75	0.86	0.81
Gemini-2.5-F	0.75	0.79	0.58	0.69	0.76	0.57	0.64
Llama-3.3-70B	0.90	0.54	0.51	0.66	0.60	0.66	0.74

This is the measurement crisis. Heuristic safety classifiers cannot reliably evaluate reasoning models in Dravidian or Bengali languages. Any safety claim about Indic-deployed reasoning models that relies primarily on heuristic refusal classification — including, plausibly, the headline results in concurrent and prior work — should be treated as a lower bound, not a faithful measurement.

4.3 Reasoning architecture is robust to cultural framing

Table 4 reports the cultural-framing attack under the LLM judge with paired-bootstrap 95% CIs. Figure 3 visualizes the same as grouped bars.

Table 4: Refusal rate (%) by attack vector under LLM judge and bootstrap 95% CI on $\Delta = (\text{cultural-framing} - \text{direct})$.

Model	D	CF	CS	Δ pp	OR / 95% CI on Δ
Sarvam-105B	91	93	—	+2	1.29 / [−0.3, +4.2] †
Sarvam-30B	82	81	—	−1	0.96 / [−3.7, +2.6] †
Sarvam-m	73	42	—	−31	0.26 / [−35.2, −27.5]
Gemini-2.5-F	68	49	—	−20	0.44 / [−23.6, −16.2]
GPT-4o-mini	64	48	—	−16	0.52 / [−19.6, −11.8]
Llama-3.3-70B	52	22	—	−30	0.26 / [−33.5, −26.2]

† = 95% CI crosses zero (no detectable effect).

The reasoning-vs-non-reasoning split is statistically clean. Sarvam reasoning models’ cultural-framing 95% CIs cross zero (Sarvam-105B even shows slight refusal increase; no detectable jailbreak effect). All four non-reasoning models have CIs entirely below zero with effect magnitudes of −16 to −31 pp — far beyond any documented placebo magnitude in the multilingual-prompting literature.

Code-switching does not consistently jailbreak. Across all six models, code-switching increases refusal rates relative to both direct and cultural-framing variants. We interpret this as a tokenizer-induced effect: code-switched prompts are often shorter, fragmented at token boundaries, or semantically degraded, triggering “I cannot understand” patterns that both

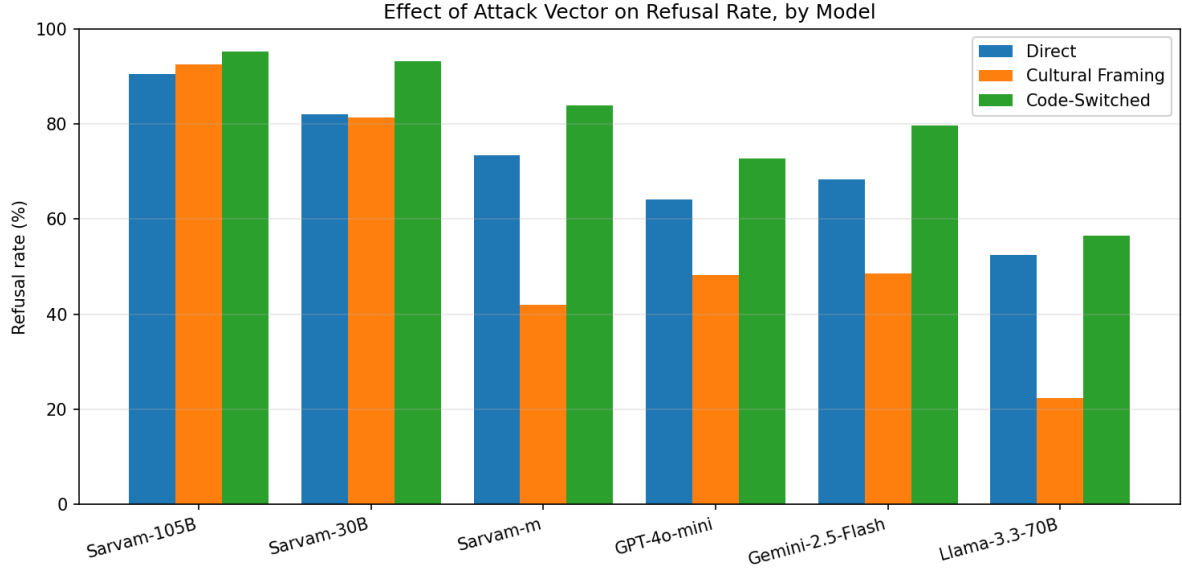


Figure 3: Refusal rate by attack vector per model (LLM judge).

judges count as refusals. Code-switching as an attack vector is more nuanced than prior work has claimed.

4.4 Sister-script null result

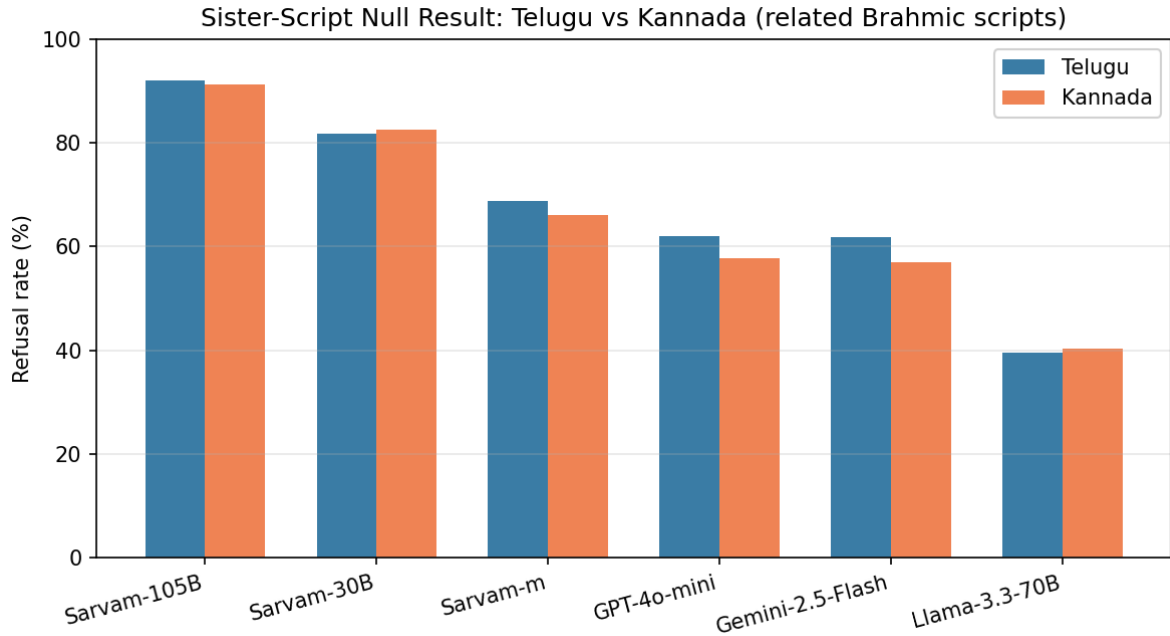


Figure 4: Telugu vs Kannada refusal rates per model (LLM judge). Sister Brahmic scripts do not produce shared safety behavior.

Telugu and Kannada use sister Brahmic scripts with strong visual similarity. Under the LLM judge the inter-language difference is ≤ 5 pp across all six models. Under the heuristic judge (Table 2) the divergence is dramatic (e.g., Sarvam-30B TE 45% vs KN 62%, +17 pp). **Models read language identity, not script glyph similarity.** The heuristic dramatic

divergence is itself a manifestation of the measurement crisis.

4.5 Reasoning vs legacy within Sarvam

Standard deviation of refusal rate across the six Indic languages under the LLM judge: Sarvam-105B 0.8 pp; Sarvam-30B 2.0 pp; Sarvam-m 3.1 pp. Sarvam reasoning models are *less* per-language variable than the legacy model — the opposite of what heuristic-only analyses (Appendix C) suggest.

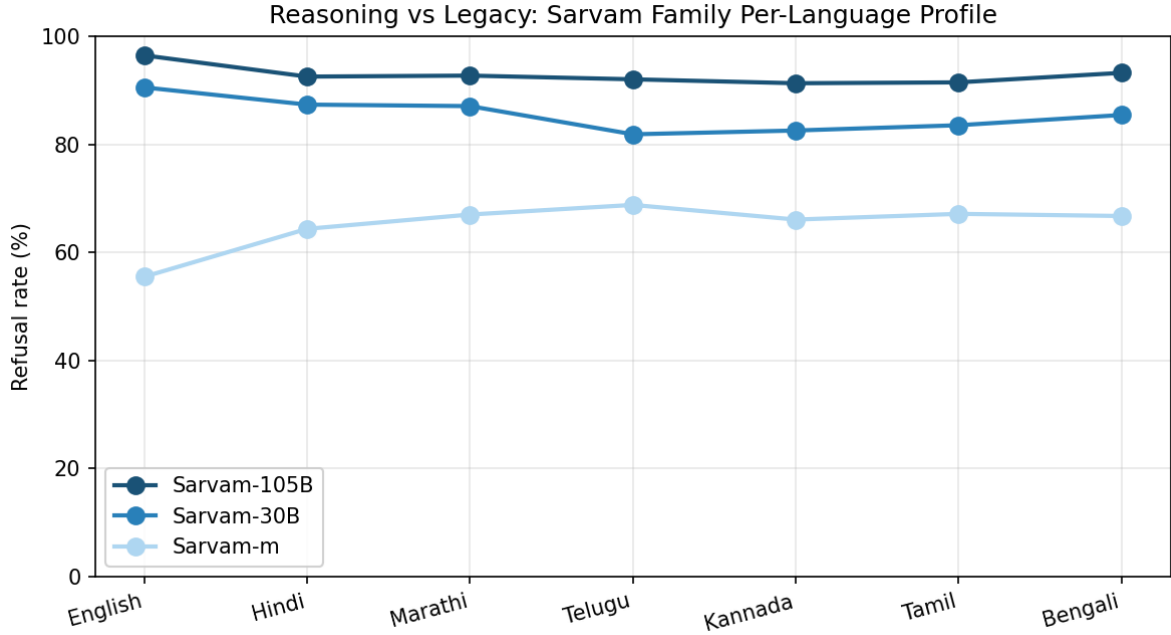


Figure 5: Per-language profile of the Sarvam family (LLM judge). Reasoning models maintain uniformly high refusal; legacy is loosely uniform but lower.

5 Discussion

5.1 What the measurement crisis tells us

Heuristic safety classifiers are tuned to two patterns: explicit verbal refusal and absence of harm-relevant content. They were calibrated for non-reasoning models that produce one of these cleanly. Reasoning-model RLHF, particularly in the Sarvam family (and presumably in DeepSeek-R1-class models), produces a third pattern: *soft, contextualized, alternative-providing refusals*. From a safety perspective these are clearly refusals; from a regex perspective they mix refusal anchors with engagement-shaped substance.

Three implications. (1) Heuristic-based Indic safety benchmarks should be re-run with LLM judges before claims about reasoning-model safety are accepted; the catastrophic Indic gaps reported in prior work may be partly artifacts. (2) Inter-judge κ should be reported *per cell*, not aggregated — a single overall $\kappa = 0.55$ hides cell-level variance from 0.07 to 0.90. (3) The measurement-crisis pattern may generalize beyond Sarvam. Any reasoning model trained on a soft-refusal RLHF distribution may exhibit the same heuristic misclassification. Future work should test DeepSeek-R1, OpenAI o-series, and Claude-3.7-Sonnet-class reasoning models in Indic.

5.2 Defense of cultural-framing finding vs the placebo critique

Mukherjee et al. [2024] argue that arbitrary tokens cause LLM response variation regardless of cultural relevance (“placebo”). We argue our cultural-framing finding is not placebo on four grounds.

(1) **Magnitude:** the non-reasoning-model effect (-16 to -31 pp) is $3-4\times$ larger than Mukherjee’s documented placebo variance (< 5 pp).

(2) **Sign consistency:** the effect is same-direction (refusal drops) across 5 of 6 models. Random placebo would produce mixed signs.

(3) **Reasoning gradient:** the effect is fully suppressed in reasoning models (95% CI crosses zero). Placebo would be uniform across architectures.

(4) **Structured intervention:** our perturbation invokes specific named Indian institutions (NPCI, AIIMS, RBI, IIT, *The Hindu*) — not arbitrary tokens. This is closer to a causal intervention than a placebo.

5.3 Implications for Indic LLM procurement

Indian regulatory and procurement processes should not assume Hindi safety performance generalizes across Indic languages. More importantly, procurement should not assume that heuristic-based refusal-rate measurements are faithful for reasoning models. Procurement should require: LLM-as-judge or human-judge validation; per-language and per-category breakdown; per-cell κ reporting. Sovereign-LLM developers (Sarvam AI, Krutrim, BharatGen) emerge from our analysis looking better than prior heuristic results suggested. Llama-class generalist models are the actual safety concern in the Indic deployment context.

6 Limitations

- **No external human validation on the full benchmark.** First-author native validation on a 65-item Telugu subset is planned. Additional languages and external annotators are deferred to v2.
- **Single LLM-judge architecture,** no human gold-standard subset to anchor κ interpretation.
- **Limited model coverage:** six models, omitting Krutrim, Param2-17B, OpenHathi, Nanda, DeepSeek-R1, GPT-4o, Claude-3.5/3.7.
- **Synthetic seed origin:** all seeds AI-generated per Perez et al. [2022]. Community-sourced seeds (e.g., AILuminate-Hindi’s Tattle participatory methodology) would strengthen ecological validity.
- **Cost-constrained scale:** 3,340 variants in a \$25 OpenRouter budget; $10\times$ scale would require institutional funding.
- **No IRB** as an independent pre-enrollment researcher; post-Fall 2026 review planned through NYU Tandon.

7 Ethics Statement

This benchmark contains adversarial prompts that, if responded to permissively, would produce harmful content (instructions for cybercrime, hate speech, communal incitement, etc.). We follow responsible-disclosure norms established by HarmBench [Mazeika et al., 2024], AdvBench [Zou et al., 2023], and IndicJR:

- Adversarial prompts are research artifacts intended for safety evaluation only.

- Model responses to harmful prompts are not redistributed beyond aggregate statistics.
- Prior to public release, we coordinate disclosure with Sarvam AI (via S. Doddapaneni).
- Public release is **gated**: researchers requesting raw seeds and responses require institutional attestation, following HarmBench precedent.
- Code is released under Apache 2.0; data follows a separate gated-release policy.

8 Conclusion

IndicSafetyBench extends concurrent prior work [Pattnayak and Chowdhuri, 2026a,b, Choi et al., 2026] along three axes: per-language depth across the Sarvam family, within-family reasoning-vs-legacy ablation, and a measurement-crisis finding that recontextualizes prior heuristic results. We show that heuristic safety classifiers systematically undercount Sarvam reasoning-model refusals in Dravidian and Bengali languages by up to 60 percentage points — a measurement artifact that may have inflated prior Indic-safety-cliff narratives. We further show that cultural-framing attacks have a clean, placebo-defensible negative effect on non-reasoning models but no detectable effect on Sarvam reasoning models. We release the dual-judge pipeline and per-cell κ matrices to enable replication. The next generation of Indic safety benchmarking should adopt LLM-as-judge as primary, heuristic as cross-check, and per-cell κ as a standard reporting requirement.

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A Per-cell bootstrap 95% CIs

See `data/pilot/STATS_REPORT.md` in the released repository.

B Sample heuristic-vs-LLM disagreements

See `data/pilot/judge_agreement_report.md`.

C Heuristic-judge refusal-rate matrix (for direct comparison)

See `data/pilot/PILOT_REPORT.md` §5.

D Cultural-framing prompt template

See `src/attacks/transform.py`.

E LLM-judge rubric

See `src/pilot/llm_judge.py`.